Cristian Necula

Senior Frontend Engineer - Web Platform & Performance

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Senior Frontend Engineer with 15 years building for the web - from founding-team SaaS to a Nasdaq-listed AI platform. Expert in Web Components, the modern web platform, React hooks internals and large-scale frontend architecture, with full-stack roots (PHP, MySQL, Varnish, Ansible). Track record of zero-downtime framework migrations, eliminating performance issues and memory leaks in apps serving millions of records, and building testing infrastructure that keeps a shared component library safely shippable. ~3,000 merged open-source pull requests since 2012, maintainer of pionjs.

EXPERIENCE

Senior Frontend Developer · Neovici (via Plumelo)

Feb 2019 – Present

Consultant through Plumelo · Cosmoz - AI financial-automation platform processing 5M+ invoices yearly, 200M+ invoice rows since 2009 (Nasdaq-listed, 2024); built on a large open-source component library

- **Led the migration of a production stack from Polymer 2 → Polymer 3 → Lit/pionjs with zero downtime**, using expert interoperability strategies to ship incrementally while features kept flowing.
- **Eliminated severe performance issues** - main-thread lagging and memory leaks - across core components of the invoicing app, and rewrote problematic components.
- **Modernized the shared component library to the latest web-platform standards** (Baseline-capable APIs), raising quality and consistency across all Cosmoz apps.
- **Built the platform's end-to-end testing system:** PollyJS-based API recordings with custom extensions for capturing Server-Sent Events (SSE) streams, wired into GitHub Actions CI.
- **Maintain pionjs**, an open-source micro-framework adding React-style hooks to Web Components, and train colleagues on the component architecture.

Full-Stack Engineer (founding team) · Widgetic / Blogvio

Jun 2015 – Feb 2019

B2B SaaS platform for embedding customizable widgets (players, feeds, forms) on third-party websites; REST API + OAuth foundation for a planned widget marketplace

- **Built the platform end to end** - PHP and Node.js backends (Express, Koa, Joi), MySQL schema, JavaScript frontend and deployment - owning everything from database to devops and caching.
- **Designed and implemented the platform's REST API with OAuth authentication**, enabling widget publishing and iframe embedding on external sites and site builders (Wix, Shopify) - the foundation for the planned third-party widget marketplace.
- **Evolved the widget frontend stack from jQuery to Svelte 1/2** as widgets grew in complexity - contributing fixes upstream to sveltejs/svelte and sveltejs/svelte-loader along the way.
- **Built the widgets' admin interfaces in React, shipped in the Wix admin UI** - letting site owners configure Widgetic widgets directly on the platform.
- **Served high-traffic embeds through a Varnish front-cache with zero downtime** on major publishers such as lefigaro.fr, handling tens of thousands of embed impressions.
- **Ran lean, repeatable production infrastructure:** services containerized with Docker (PHP, MySQL, Varnish) on three minimal DigitalOcean servers, provisioned with Ansible.

Web Developer · Vuzum

2010 – May 2015

Web agency of ~20 people building client websites

- Delivered PHP websites for small-business clients, starting with WordPress and advancing to Symfony 2, with jQuery-based frontends.

OPEN SOURCE

~3,000 open-source pull requests merged since 2012 - 2,900+ across Neovici's open-source component ecosystem (the OSS layer powering Cosmoz) and ~50 to external projects, including svelte and haunted.

pionjs · Maintainer

github.com/pionjs/pion · 127★

- Open-source micro-framework implementing React's Hooks API for Web Components; runs in production across the Cosmoz platform. Also contributed to **haunted** (2.7k★), the hooks-for-Web-Components library that pion derives from.

Cosmoz component ecosystem · Neovici

github.com/Neovici

- Long-term contributor and maintainer across the org's open-source ecosystem - commits in **62 of its 80 public repositories** (5,000+ commits) on shared components and libraries (cosmoz-omnitable, cosmoz-router, cosmoz-form, and more) published on npm.

External contributions

- **Early Svelte contributor (2017)** - merged PRs into sveltejs/svelte and sveltejs/svelte-loader while building production widgets with Svelte 1/2 at Widgetic.
- Merged PRs into inline-source, GitGutter, JMSDebuggingBundle and other community projects.

Personal projects

- **knowledge-mcp** - TypeScript MCP server implementing a brain-inspired memory model (knowledge decay, associative linking, knowledge-graph visualization) for agent tooling.
- **Spelling Bee** - educational browser game for kids built with Phaser 4 and TypeScript, including in-browser text-to-speech via kokoro-js.

SKILLS

Languages	JavaScript (ES modules), TypeScript, PHP, HTML, CSS
Frontend	Web Components, Lit / lit-html, React (app development + hooks internals via pionjs), modern CSS, performance profiling, Baseline web APIs
Backend	Node.js (Express, Koa, Joi), REST API design, OAuth, PHP / Symfony, MySQL
Infra & DevOps	Docker, Varnish, Ansible, DigitalOcean, GitHub Actions, CI/CD
Tooling & Testing	Vite, PollyJS (API recording), SSE streaming, end-to-end testing
Practices	Zero-downtime migrations, component library architecture, mentoring

EDUCATION & LANGUAGES

BSc Computer Science · Alexandru Ioan Cuza University, Iași — 2007 – 2010

"Grigore Moisil" Computer Science Highschool · Iași — first HTML websites, first paid web development work at age 16 — 2003 – 2007

Romanian (native) · English (fluent)